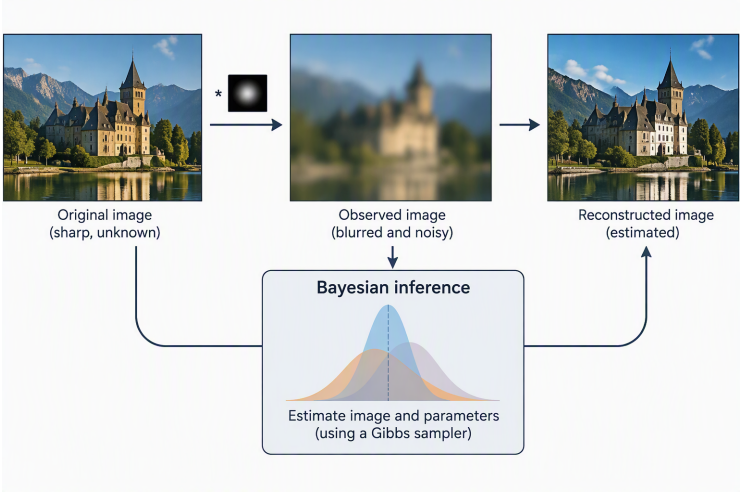
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Estimation and identification

Au-512, 5th year SET, IPSA

Practical works 2: image deblurring using Gibbs Sampling

Image deblurring is a classical inverse problem in signal and image processing. Due to optical imperfections, motion, or sensor limitations, the acquired image is often degraded and does not accurately represent the original scene. In this laboratory session, we consider a Bayesian approach to image restoration. To estimate the image, as well as the noise and regularization levels, we adopt a hierarchical Bayesian model and implement a Gibbs sampler. This approach allows the hyperparameters to be estimated automatically from the observed data, leading to an unsupervised image restoration method. The objective of this laboratory session is to derive the conditional distributions involved in the model, implement the Gibbs sampler, and analyze the reconstructed image.



1 Observation Model

We consider the following degradation model:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{e},$$

where $\mathbf{x}$ is the unknown sharp image, $\mathbf{y}$ is the observed blurry image, $\mathbf{H}$ is a blur operator, $\mathbf{e} \sim \mathcal{N}(0, \tau_e^{-1} I)$

To promote smooth solutions, we use the following prior model:

$$p(\mathbf{x}|\tau_x) \propto \exp\left(-\frac{\tau_x}{2}\|\mathbf{D}\mathbf{x}\|^2\right),$$

where $\mathbf{D}$ denotes the finite-difference operator.

Question 1
Explain the physical meaning of τ_e and τ_x, and $\mathbf{D}$.

2 Analysis of the Observed Image

Open the file "main.m", which contains the skeleton of the Gibbs sampler to be completed throughout this practical work. Load the provided image Y and the true image X:

```
%True image
figure()
imshow(X,[])
%Acquired image
figure()
imshow(Y,[]
```

Question 2
Describe the visible degradation of the image.

Question 3
Display the blur kernel:

```
figure;
imagesc(psf);
colorbar;
axis image;
```

What is the role of this kernel?

3 Fourier-Domain Representation

Since the blur operator corresponds to a convolution, it is convenient to express the model in the Fourier domain. Applying the discrete Fourier transform (DFT) to the observation model yields

$$\widehat{\mathbf{y}} = \widehat{\mathbf{H}} \odot \widehat{\mathbf{x}} + \widehat{\mathbf{e}},$$

where $\odot$ denotes the Hadamard (element-wise) product. $\widehat{\mathbf{x}}$: Fourier coefficients of the unknown image, $\widehat{\mathbf{y}}$: Fourier coefficients of the observed image, $\widehat{\mathbf{H}}$: transfer function associated with the blur kernel, $\widehat{\mathbf{e}}$: Fourier transform of the additive Gaussian noise. Similarly, the Gaussian prior can be written in the Fourier domain as

$$p(\widehat{\mathbf{x}}|\tau_x) \propto \exp\left(-\frac{\tau_x}{2}\|\widehat{\mathbf{D}} \odot \widehat{\mathbf{x}}\|^2\right),$$

where $\widehat{D}$ denotes the Fourier transform of the finite-difference operator.

Since the Fourier transform diagonalizes convolution operators, the operator

$$\mathbf{D}^\top \mathbf{D} = \mathbf{D}_x^\top \mathbf{D}_x + D_y^\top D_y$$

is represented in the Fourier domain by

$$\tilde{\mathbf{D}} = |\widehat{\mathbf{D}}_x|^2 + |\widehat{\mathbf{D}}_y|^2.$$

where $\widehat{\mathbf{D}}_x = \begin{bmatrix} 1 & -1 \end{bmatrix}$ computes horizontal differences, and $\widehat{\mathbf{D}}_y = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$ vertical differences.

Question 4

Implement $\tilde{\mathbf{D}}$ in the Fourier domain in the function "Fourier-D.m'. Explain why it will be useful in the following,

4 Gibbs sampler

The posterior distribution is given by

$$p(x, \tau_e, \tau_x | \mathbf{y}) \propto p(\mathbf{y}|\mathbf{x}, \tau_e) p(\mathbf{x}|\tau_x) p(\tau_e) p(\tau_x).$$

The Gibbs sampler alternates between:

1. $\tau_e \sim p(\tau_e | \mathbf{x}, \mathbf{y})$,
2. $\tau_x \sim p(\tau_x | \mathbf{x}, \mathbf{y})$,
3. $\mathbf{x} \sim p(\mathbf{x} | \tau_e, \tau_x, \mathbf{y})$.

and we aim to complete the Gibbs loop:

```
for it = 1:nIter

% Sample tau_e

...

% Sample tau_x

...

% Sample x

...

end
```

Throughout this laboratory, we assume $\tau_e \sim \Gamma(\alpha_{e0}, \beta_{e0})$ and $\tau_x \sim \Gamma(\alpha_{x0}, \beta_{x0})$.

4.1 Conditional Distribution of τ_e

Starting from Bayes' rule, show that

$$p(\tau_e|\mathbf{x},\mathbf{y}) = \Gamma(\tau_e;\alpha_e,\beta_e),$$

with

$$\alpha_e = \alpha_{e0} + \frac{N}{2} \tag{1}$$

$$\beta_e = \beta_{e0} + \frac{1}{2}\|\mathbf{y}-\mathbf{H}\mathbf{x}\|^2. \tag{2}$$

Question 5
Now, complete the following MATLAB code:

```
% Sample tau_e


alpha_e = ..............................;

beta_e = ..............................;

tau_e =
rndgamma_shape_rate(alpha_e,beta_e);
```

4.2 Conditional Distribution of τ_x

Starting from the prior model,

$$p(\mathbf{x}|\tau_x) \propto \tau_x^{N/2} \exp\left(-\frac{\tau_x}{2}\|\mathbf{D}\mathbf{x}\|^2\right),$$

show that

$$\tau_x|\mathbf{x},\mathbf{y} = \Gamma(\alpha_x,\beta_x).$$

Question 6
Show that

$$\alpha_x = \alpha_{x0} + \frac{N}{2}.$$

and determine the expression of β_x

Complete the following MATLAB code:

```
xnormD2 = ..........................;

alpha_x = ..........................;

beta_x = ..........................;

tau_x =
rndgamma_shape_rate(alpha_x,beta_x);
```

4.3 Conditional Distribution of the Image

The conditional distribution

$$\mathbf{x}|\tau_e, \tau_x, \mathbf{y}$$

is Gaussian with mean $\boldsymbol{\mu} = \left(\tau_e\ , \mathbf{H}\mathbf{H}^\top + \tau_x \mathbf{D}^T\mathbf{D}\right)^{-1} \tau_e \mathbf{H}\mathbf{y}$
In the Fourier domain, its mean is

$$\widehat{\boldsymbol{\mu}} = \left(\tau_e \widehat{\mathbf{H}} \odot \widehat{\mathbf{y}}\right) \oslash \left(\tau_e \left|\widehat{\mathbf{H}}\right|^2 + \tau_x \tilde{\mathbf{D}}\right)$$

and its covariance is

$$\widehat{\boldsymbol{\Sigma}} = \tau_e |\widehat{\mathbf{H}}|^2 + \tau_x \tilde{\mathbf{D}}$$

where the division is element wise.

Question 8
Explain why working in the Fourier domain greatly reduces the computational cost.

Question 9
Complete the computation of $\widehat{\boldsymbol{\mu}}$:

```
num=
..................................;

denom =
..................................;

muhat =num ./denom
..................................;
```

Question 10
Complete the Gaussian sampling step :)

```
U = fft2(randn(M,N))/sqrt(Npix);%Scaled Gaussian white noise in Fourier domain

X =...
x = real(ifft2(X));
```

5 Convergence Study

Store the generated samples:

```
tauE(it)=tau_e;
tauX(it)=tau_x;
```

Plot:

```
figure;
subplot(2,1,1);
plot(tauE);

subplot(2,1,2);
plot(tauX);
```

Question 12
What is the purpose of the burn-in period?

Question 13
After how many iterations do the chains appear stable?

6 Posterior Mean Estimator

The Bayesian estimator of the image is defined by

$$\hat{\mathbf{x}} = \frac{1}{K} \sum_{k=1}^{K} \mathbf{x}^{(k)}.$$

Question 14
Explain

```
if it > BurnIn &&  mod(it - BurnIn, Thin) =  0
            kidx = kidx + 1; kept{kidx} = x;
        end
```

and complete

```
Xbar = zeros(M,N);

for k=1:kidx

Xbar =
.............................;

end

Xbar =
.............................;
```

7 Discussion

Question 15

Compare the observed image and the reconstructed image.

Question 16

What is the advantage of estimating τ_e and τ_x automatically instead of fixing them manually?

Question 17

Why can this method be considered as an unsupervised Bayesian deconvolution method?